Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

1. Calculate the heat required to raise 1 L of water from 30 Â°C to 60 Â°C:

Volume of 1 L = 1,000 g (1 g/mL for water)

Q = mcÎĶT (heat = mass ÃĹ specific heat ÃĹ temperature change)

ÎĶT = 60 Â°C - 30 Â°C = 30 Â°C

Q = (1,000 g) ÃĹ (1 cal/g Â°C) ÃĹ (30 Â°C) = 30,000 cal = 30 Kcal

2. Calculate the heat required to raise 1 kg (1,000 g) of aluminum from 30 Â°C to 60 Â°C:

Q = (1,000 g) ÃĹ (0.215 cal/g Â°C) ÃĹ (30 Â°C) = 6,450 cal = 6.45 Kcal

So, the answer is (G). 30 Kcal, 6.45 Kcal

</think><answer> G